This line of inquiry will shortly take us to world-shattering collisions, to two quite different perils for our species, and to a reason—beyond those already described—that we must, for our very survival, be out there among the planets.

We now know that the rings (emphatically plural) of Saturn are a vast horde of tiny ice worlds, each on its separate orbit, each bound to Saturn by the giant planet's gravity. In size, these worldlets range from particles of fine dust to houses. None is big enough to photograph even from close flybys. Spaced out in an exquisite set of fine concentric circles, something like the grooves on a phonograph record (which in reality make, of course, a spiral), the rings were first revealed in their true majesty by the two *Voyager* spacecraft in their 1980/81 flybys. In our century, the Art Deco rings of Saturn have become an icon of the future.

At a scientific meeting in the late 1960s, I was asked to summarize the outstanding problems in planetary science. One, I suggested, was the question of why, of all the planets, only Saturn had rings. This, *Voyager* discovered, is a nonquestion. *All four* giant planets in our Solar System—Jupiter, Saturn, Uranus, and Neptune—in fact have rings. But no one knew it then.

Each ring system has distinctive features. Jupiter's is tenuous and made mainly of dark, very small particles. The bright rings of Saturn are composed mainly of frozen water; there are thousands of separate rings here, some twisted, with strange, dusky, spokelike markings forming and dissipating. The dark rings of Uranus seem to be composed of elemental carbon and organic molecules—something like charcoal or chimney soot; Uranus has nine main rings, a few of which sometime seem to "breathe," expanding and contracting. Neptune's rings are the most tenuous of all, varying so much in thickness that, when detected from Earth, they appear only as arcs and incomplete circles. A number of rings seem to be maintained by the gravitational tugs of two shepherd moons, one a little nearer and the other a little farther from the planet than the ring. Each ring system displays its own, appropriately unearthly, beauty.